

4.3 How the derivative affects the shape of the graph (Pages 293 – 304 in the text book)

PART 1:

Increasing / Decreasing Test

- (a) If $f'(x) > 0$ on an interval, then f is increasing on that interval
- (b) If $f'(x) < 0$ on an interval, then f is decreasing in that interval

First Derivative Test

If $f'(x)$ changes from positive to negative at c then f has a local maximum at c .
If $f'(x)$ changes from negative to positive at c then f has a local minimum at c .

PART 2: Concavity

Concavity test

- (a) If $f''(x) > 0$ on an interval I then f is concave upward on I
- (b) If $f''(x) < 0$ on an interval I then f is concave downward on I

Second derivative test

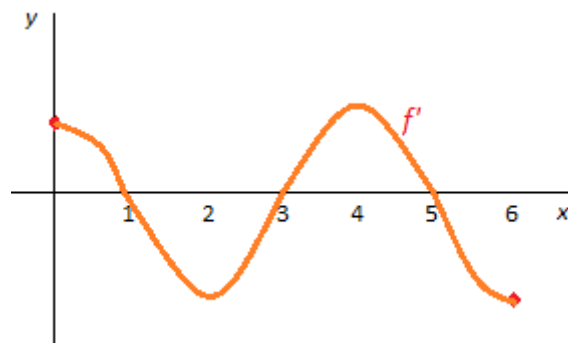
Suppose f'' is continuous near c . Then the following hold:

- (a) If $f'(c) = 0$ and $f''(c) > 0$, then f has a local minimum at c
- (b) If $f'(c) = 0$ and $f''(c) < 0$, then f has a local maximum at c

PART 3: Application

Example 1

The sketch shows the graph of the derivative f' of a function. Study the graph and then answer the following questions:



- (a) On what interval(s) is f increasing?

Remember: A function f is increasing where $f'(x) > 0$, that is where the value of the first derivative is positive.

In this case you can see that the values of f' are positive (above the x-axis) on the intervals $(0, 1)$ and $(3, 5)$. Thus, f is increasing on the intervals $(0, 1)$ and $(3, 5)$.

- (b) At what values of x does f possibly have a local maximum or a local minimum?

Remember: The critical numbers are the numbers c where either $f'(c)$ does not exist or where $f'(c) = 0$.

In this case the derivative exists on the interval $[0, 6]$ and so the only critical numbers are where $f'(c) = 0$, that is where the graph of f' intersects the x-axis.

So f possibly has local extremes at $x = 1, x = 3$ and $x = 5$.

(c) On which intervals is f concave upwards or concave downwards?

We want to use the concavity test, so we first need to make some conclusions about the sign of the second derivative. For this we will use the increasing/decreasing test:

Note that the graph shows that f' is increasing on the interval $(2, 4)$. This means that the derivative of f' must be positive on $(2, 4)$. But the derivative of f' is f'' and so we have that $f''(x) > 0$ on $(2, 4)$. Thus, by the concavity test, f is concave upwards on $(2, 4)$.

Likewise, since f' is decreasing on $(0, 2)$ and on $(4, 6)$, it follows that $f''(x) < 0$ on $(0, 2)$ and on $(4, 6)$. Thus, by the concavity test, f is concave downwards on $(0, 2)$ and on $(4, 6)$.

(d) Find the x-coordinates of the inflection points of f .

Remember: The inflection points are the points where the concavity changes. We know from (c) that f is concave downwards on $(0, 2)$ but concave upwards on $(2, 4)$. Thus, there is an inflection point at $x = 2$. Similarly, since f is concave upwards on $(2, 4)$ but concave downwards on $(4, 6)$. Thus there is also an inflection point at $x = 4$.

Example 2

Let f be a function that satisfies the following conditions:

(a) $f'(x)$ exists everywhere and $f'(x) = 0$ only when $x = 0, x = 2/3$ or $x = 1$.

(b) $f''(0) = 0$; $f''(2/3) > 0$ and $f''(1) = 0$

(c) $f'(x) > 0$ on the intervals $(-\infty, 0)$; $(2/3, 1)$ and $(1, \infty)$; $f'(x) < 0$ on the interval $(0, 2/3)$.

Discuss the local extremes of the function f .

Solution

It follows from (a) that the only critical numbers of f are $x = 0, x = 2/3$ and $x = 1$.

The fact that $f'(0) = 0 = f''(0)$ and $f'(1) = 0 = f''(1)$ means that we cannot use the second derivative test at $x = 0$ and $x = 1$.

However, since $f'(2/3) = 0$ and $f''(2/3) > 0$, we can conclude by the second derivative test that f has a local minimum at $x = 2/3$.

It is given that $f'(x) > 0$ on the intervals $(-\infty, 0)$ and that $f'(x) < 0$ on the interval $(0, 2/3)$.

Thus, $f'(x)$ changes from positive to negative at 0 and so, by the first derivative test, f has a local maximum at 0.

It is also given that $f'(x) > 0$ on the intervals $(2/3, 1)$ and $(1, \infty)$. Thus the sign of $f'(x)$ does not change at $x = 1$ and so the first derivative test does not yield an outcome here.

In this case, we do not have enough information to determine the nature of the critical number $x = 1$.